

WQEye: A Python-based Software Aided by Google Earth Engine for Machine Learning-based Retrieval of Water Quality Parameters from Sentinel-2 and Landsat-8/9 Remote Sensing Data

1. Step-by-Step Tutorial

After successful installation, WQEye is launched automatically in your default web browser.

1.1. User Interface (Figure 1)

The left-hand panel displays the main modules of WQEye, corresponding to the key steps in the workflow: Data Loader, RS Sampling, Matching, Preprocessing, Machine Learning, and Export. Each module follows the structure described in detail in the paper.

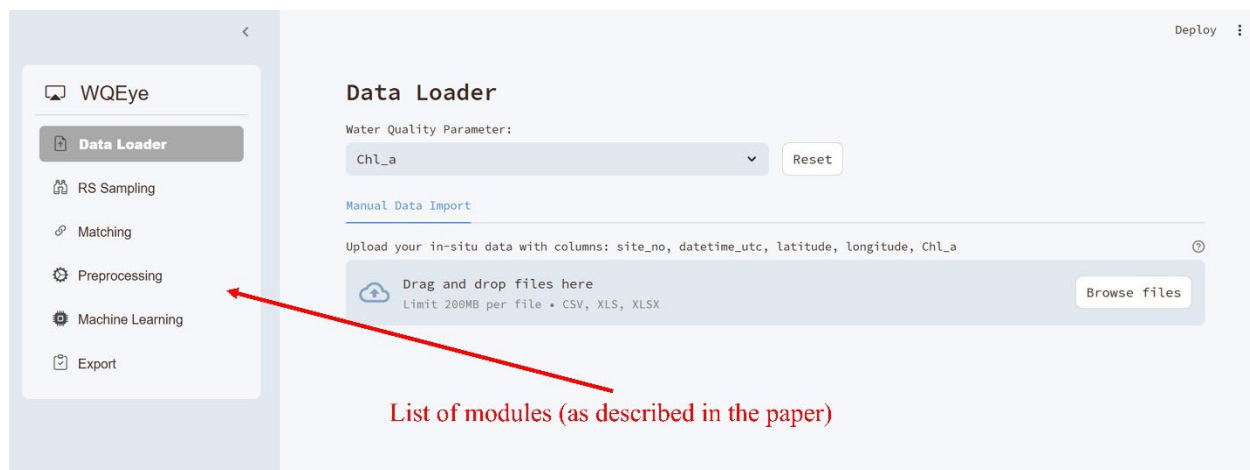


Figure 1. User Interface.

1.2. Data Loader Module

The workflow begins with the Data Loader Module, where users upload in-situ Water Quality Parameter (WQP) data. Uploaded datasets must follow the standardized format outlined below:

Column Name	Description
site_no	Unique identifier for each sampling station
latitude	Geographic latitude in decimal degrees (e.g., 36.224578)
longitude	Geographic longitude in decimal degrees (e.g., 56.542136)
<i>WQP</i>	Target water quality parameter
datetime_utc	Date and time of measurement in UTC (format: MM/DD/YYYY HH:MM)

Note:

- Latitude and longitude must be in decimal degrees.
- WQP values must use consistent units across all records.
- datetime_utc must follow the exact format MM/DD/YYYY HH:MM.

Figure 2 illustrates correctly formatted in-situ data for the Chl-a parameter at a single station St1. Users may provide a single file containing all stations (ensuring site_no is unique per station) or multiple files, each corresponding to a different station. Accepted file formats are: .CSV, .XLS, .XLSX. A sample dataset is available in the software repository for reference.

site_no	latitude	longitude	Chl-a	datetime_utc
St1	36.22	56.54	45.1	6/29/2018 13:45
St1	36.22	56.54	44.2	6/29/2018 14:00
St1	36.22	56.54	43.8	6/29/2018 14:15
St1	36.22	56.54	43.5	6/29/2018 14:30
St1	36.22	56.54	44.4	6/29/2018 14:45
St1	36.22	56.54	44.3	6/29/2018 15:00
St1	36.22	56.54	46.2	6/29/2018 15:15
St1	36.22	56.54	44.9	6/29/2018 15:30
St1	36.22	56.54	44.7	6/29/2018 15:45
St1	36.22	56.54	44.3	6/29/2018 16:00
St1	36.22	56.54	44.2	6/29/2018 16:15
St1	36.22	56.54	44.1	6/29/2018 16:30
St1	36.22	56.54	44	6/29/2018 16:45
St1	36.22	56.54	44.5	6/29/2018 17:00
St1	36.22	56.54	44.3	6/29/2018 17:15

Figure 2. Correctly formatted in-situ data for the Chl-a parameter.

Figure 3 demonstrates how to upload and manage in-situ WQP data within the Data Loader Module of WQEye. The first step in the process is to select the specific WQP the user intends to work on. Then, the user proceeds to upload the corresponding in-situ measurement data. This is accomplished by clicking the Browse files button, which allows the user to select one or more data files from their local system. After uploading the data, WQEye provides the option to preview the contents of the uploaded file(s). This is done by enabling the Show data preview toggle. When activated, a table is displayed, showing an overview of the dataset, including its total size and a sample of the records. This preview step allows the user to quickly verify that the data format and content are correct. If the uploaded data appears satisfactory, the user can click the Confirm and Save Data button to finalize the upload and proceed to the next module in the WQEye workflow. In case of any issues with the uploaded files or if the user wishes to restart the process, the Reset button can be used. This option removes all previously uploaded files and allows the process to begin from the start. In Figure 3, a unified file containing data from three monitoring stations has been uploaded for turbidity mapping. The dataset is reviewed using the preview option, and once confirmed to be correct, the user proceeds to the next step.

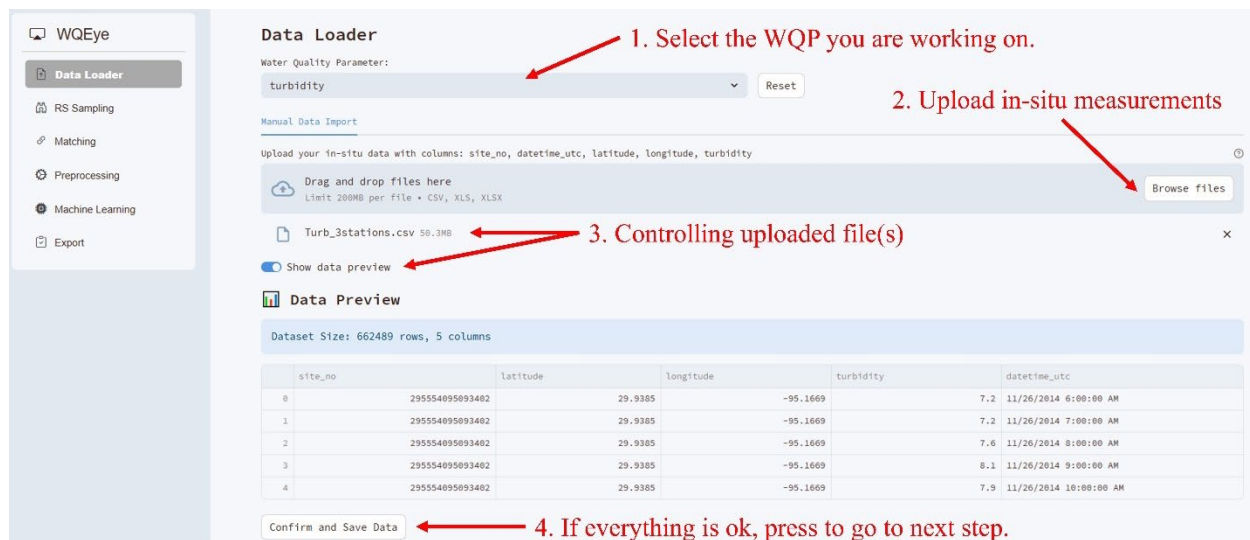


Figure 3. Data Loader module.

1.3. RS sampling Module

Once the in-situ data has been successfully uploaded, the user proceeds to the RS Sampling Module (Figure 4), where remote sensing data are prepared for the analysis. At the top of this interface, WQEye provides a summary of the uploaded in-situ data. This includes the date of the first and last in-situ measurements, as well as the total number of unique monitoring stations identified in the uploaded dataset. This overview allows the user to verify that the correct dataset is being used for subsequent remote sensing data extraction. The next step is to select the remote sensing dataset to be used for sampling. WQEye currently supports Sentinel-2, Landsat-8, Landsat-9, and a combined Landsat-8 and Landsat-9 dataset. In Figure 4, Sentinel-2 has been selected as the source of satellite data. In addition to selecting the dataset, users have the option to define a global cloud coverage threshold and a buffer distance (please refer to the paper for further details). Based on the latitude and longitude information provided during the data upload, WQEye automatically identifies all unique monitoring stations. These are displayed along with their corresponding geographic coordinates. The user can then select which stations to include in the analysis by checking the relevant boxes. In Figure 4, all stations have been selected for further processing. Once the selections are complete, the user clicks the Submit button to confirm their choices. The software will then automatically extract satellite reflectance values for selected locations.

Remote Sensing Data Overview

Summary of Data

Data Overview

First in-situ measurement: 2013-03-18

Last in-situ measurement: 2025-02-11

Number of unique sites: 3

Select Sensor

☒ Sentinel-2 ☐ Landsat-8 ☐ Landsat-9

Global Cloud Coverage (%) 20% 0% 100%

Buffer Distance (m) 250 - +

Stations to use further: 3

Station Code	Latitude	Longitude	
2955548950934	29.9385388154	-95.166944176	☒
2958260950822	29.9741338626	-95.139616277	☒
3000320950805	30.0090176295	-95.134755842	☒

☒ Submit

1. Select RS data, global cloud coverage and buffer distance for cloud filtering (for further details, refer to the paper)

2. Which stations the user finally wants to include (checked stations will be included).

3. Press submit.

Figure 4. RS sampling module.

1.4. Matching module

In this module (Figure 5), in-situ and satellite observations are paired based on their temporal proximity. The user begins by entering a time difference threshold, expressed in seconds. This threshold defines the maximum allowable time gap between an in-situ observation and the corresponding satellite image acquisition. Once the threshold is set, the user initiates the matching process by clicking the Run Matching button. When the matching process is completed successfully, the software displays a confirmation message indicating that all sites have been processed. At this stage, the user has the option to download the matched dataset by clicking the Download Matched Data button. The matched data can be used in other software platforms or integrated into the user's own custom code for further analysis if desired.

Matching

Enter the matching threshold (in seconds)

3600 - +

Processing station 3/3 (site_no: 300032095080501)

☒ Matching completed for all sites.

1. Setting the time difference threshold between in-situ and satellite observations

2. Press 'Run Matching'

3. (Optional) Once the process is done successfully, the user can download matched data to use in other software, platforms, and their own codes to work on.

Figure 5. Matching module.

WQEye generates a matched dataset, which is provided in a .csv format for convenient use in further analyses or model development (Figure 6). Each record in the matched dataset contains a comprehensive set of information that links satellite observations to the corresponding in-situ water quality measurements. Specifically, the file includes the spectral band values extracted from the selected remote sensing dataset (Sentinel-2 bands are shown in Figure 6). In addition to the spectral information, the file contains both the image acquisition time and the in-situ measurement time, expressed in Coordinated Universal Time (UTC). The dataset also includes the site number (site_no), which corresponds to the station identifiers provided by the user during the initial data upload. Further metadata are provided in the form of the spacecraft platform name, indicating the specific satellite from which the data were acquired (e.g., Sentinel-2A, Sentinel-2B, or Sentinel-2C). The tile_name column specifies the image scene (tile) from which the reflectance values were extracted, based on the Military Grid Reference System (MGRS) for Sentinel-2 data. Finally, the file contains the measured in-situ water quality parameter value for each record. In Figure 6, turbidity is the parameter of interest, and its corresponding values are listed in the final column. This matched dataset provides all the essential information required for model development, statistical analysis, or integration into the user's own workflows. The structured and transparent format ensures that users can readily apply the data for machine learning tasks or for validation purposes in other platforms and software environments.

B1	B11	B12	B2	B3	B4	B5	B6	B7	B8	B8A	B9	ImageAcquisition_time	data_utc	site_no	spacecraft_name	tile_name	turbidity
0.0466	0.0054	0.0053	0.0747	0.1064	0.129	0.1309	0.0617	0.0587	0.0515	0.0323	0.0039	12/20/2018 17:05	12/20/2018 17:00	2.95554E+14	Sentinel-2B	15RTP	73.6
0.0449	0.0049	0.0046	0.0771	0.111	0.1342	0.1333	0.0637	0.0624	0.0515	0.0346	0.0008	12/23/2018 17:15	12/23/2018 17:00	2.95554E+14	Sentinel-2B	15RTP	91.7
0.0792	0.0563	0.0535	0.0814	0.1048	0.1092	0.1211	0.0829	0.0839	0.0573	0.072	0.089	3/20/2019 17:05	3/20/2019 17:00	2.95554E+14	Sentinel-2B	15RTP	63.2
0.0533	0.0083	0.0054	0.0755	0.1082	0.1104	0.1093	0.0453	0.044	0.0379	0.0257	0.0107	4/9/2019 17:05	4/9/2019 17:00	2.95554E+14	Sentinel-2B	15RTP	59.5
0.0603	0.0143	0.0124	0.0776	0.111	0.1074	0.1113	0.0514	0.0517	0.0421	0.0303	0.0174	4/14/2019 17:05	4/14/2019 17:00	2.95554E+14	Sentinel-2A	15RTP	55.7
0.0563	0.0223	0.0216	0.0833	0.1138	0.1058	0.1081	0.051	0.05	0.0477	0.0369	0.027	4/19/2019 17:05	4/19/2019 17:00	2.95554E+14	Sentinel-2B	15RTP	49.3
0.1243	0.0921	0.0842	0.1556	0.1874	0.1708	0.179	0.1207	0.1214	0.115	0.1023	0.0967	4/22/2019 17:15	4/22/2019 17:00	2.95554E+14	Sentinel-2B	15RTP	44.8
0.1406	0.1035	0.0913	0.1838	0.2124	0.1848	0.2051	0.129	0.1277	0.1278	0.1095	0.0869	5/2/2019 17:15	5/2/2019 17:00	2.95554E+14	Sentinel-2B	15RTP	35.9
0.0574	0.0213	0.0183	0.065	0.0883	0.1094	0.1163	0.0653	0.0675	0.0583	0.0449	0.0248	5/14/2019 17:05	5/14/2019 17:00	2.95554E+14	Sentinel-2A	15RTP	73
0.1311	0.111	0.1002	0.1472	0.1586	0.1776	0.1876	0.14	0.1374	0.1288	0.1236	0.1046	5/22/2019 17:15	5/22/2019 17:00	2.95554E+14	Sentinel-2B	15RTP	38.4
0.2231	0.2301	0.197	0.2714	0.2978	0.3072	0.3128	0.2566	0.2478	0.2481	0.2363	0.1924	6/6/2019 17:15	6/6/2019 17:00	2.95554E+14	Sentinel-2A	15RTP	28.6

Figure 6. An example of matched data.

1.5. Preprocessing module

The Preprocessing module (Figure 7) contains three submodules for managing and preparing data for model development. In the first submodule, Load & Inspect Data, the matched dataset from the previous step is automatically displayed. If the user already has matched data from previous runs or external sources, they can also upload it here, provided it follows the same format as the generated matched file.

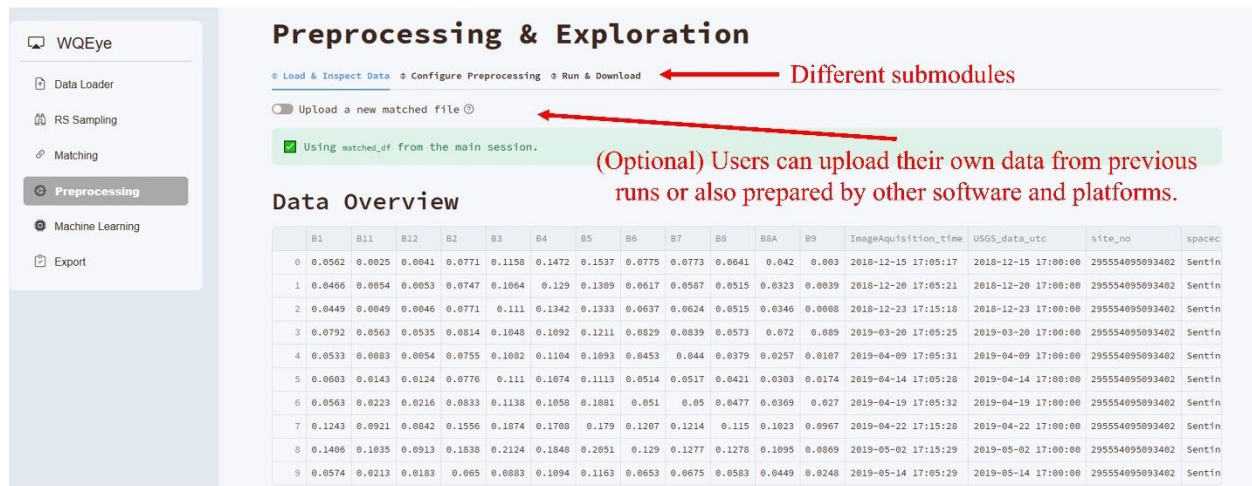


Figure 7. Preprocessing module (overview).

In the Configure Preprocessing submodule (Figure 8), the user selects which remote sensing spectral bands to include in the analysis and defines the proportion of training and testing data. During this step, the data are also preprocessed using the LogScale transformation, as described in detail in the paper. Additionally, the user can view the distribution of the matched water quality parameter values to assess data variability before proceeding.

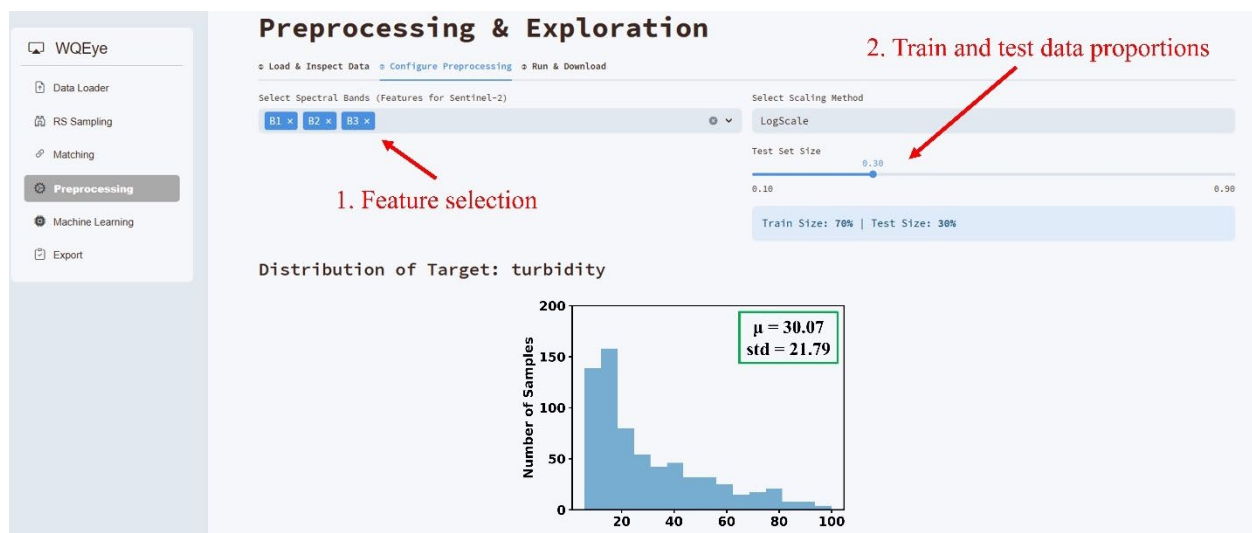


Figure 8. Preprocessing module (Configure Preprocessing).

In the Run & Download submodule (Figure 9), the user is first shown an overview of the selected preprocessing configuration, including the target variable, chosen spectral bands, scaling method, and the train-test split. After reviewing this information, the user clicks the Generate Processed

Datasets button to apply the transformations. Once completed, a download option becomes available, allowing the user to export training and testing datasets for use in other software, platforms, or their own custom code.

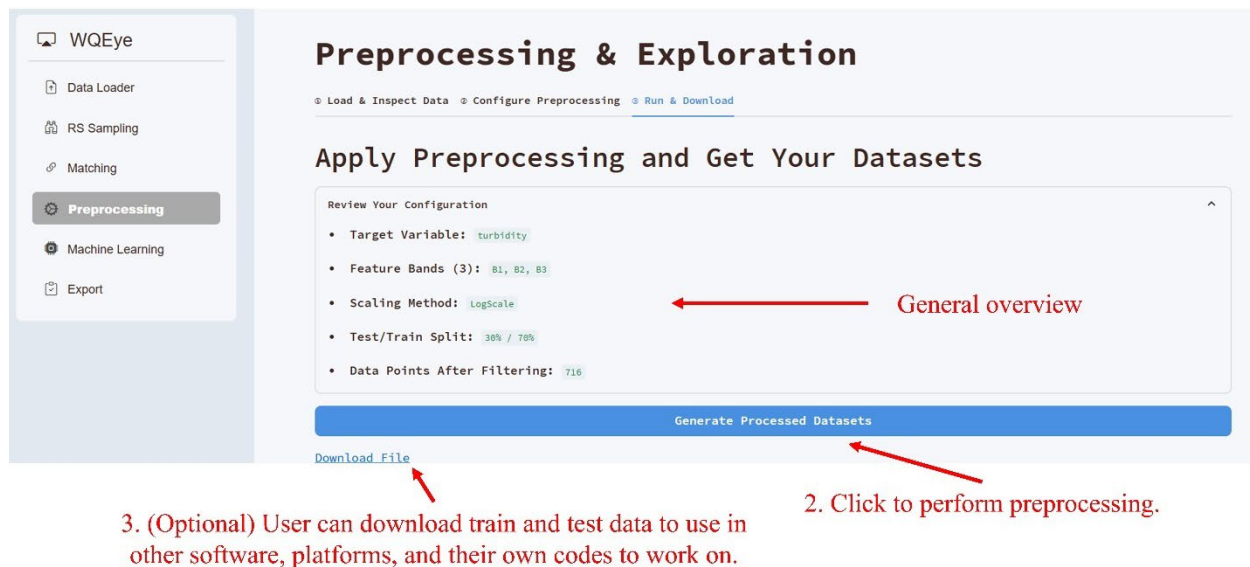


Figure 9. Preprocessing module (Run and Download).

The downloaded files from the preprocessing step include the training and testing datasets as well as a .pkl file containing the transformation parameters. This .pkl file allows users to reverse the transformations and convert the data back to its original scale if needed. The training and testing datasets contain the selected remote sensing spectral bands, in this example, bands B1, B2, and B3, and the transformed target water quality parameter values, ready for model development.

B1	B2	B3	target
0.047082	0.06729	0.098102	0.67298
0.006631	0.004338	0.025034	0.168417
0.047698	0.063778	0.108948	0.772791
0.018155	0.013885	0.034578	0.191036
0.347298	0.487336	0.520993	0.028564
0.011962	0.016619	0.038991	0.339289
0.09324	0.0566	0.084977	0.280199
0.054215	0.068918	0.107676	0.668548

Figure 10. A sample training data.

1.6. ML module

The ML module (Figure 11) contains several submodules that guide the user through the model development. In the Model Selection submodule, the user chooses one of three available machine learning models: Random Forest, Artificial Neural Network, or Kolmogorov-Arnold Network. The module also provides an overview of the dataset, including the number of training and testing samples, as well as the number of selected input features. If users already have their own prepared training and testing datasets from previous runs or external tools, they can upload them directly for use in this module. In the example shown, the Random Forest model has been selected for training.

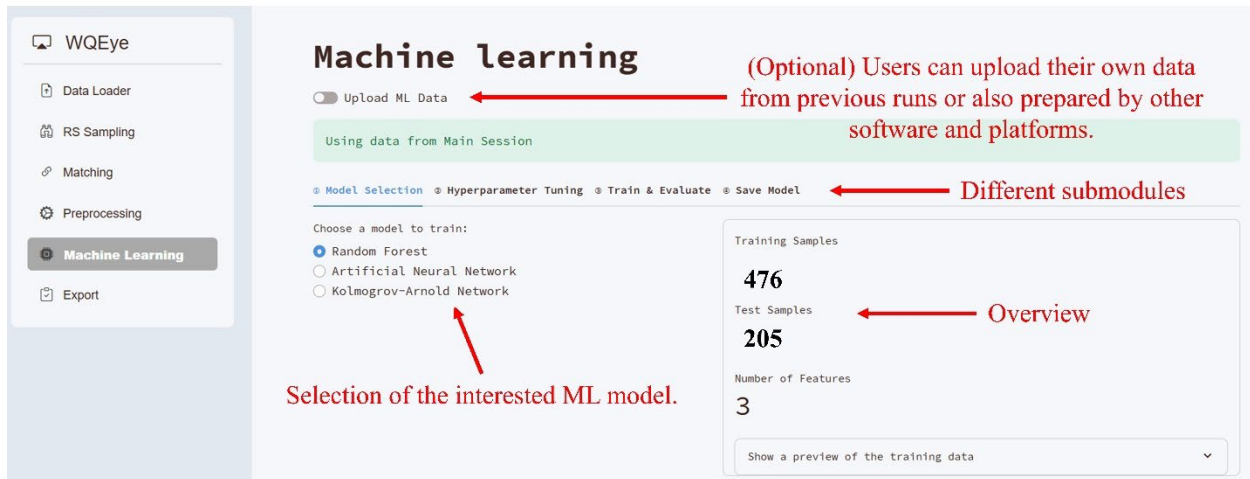


Figure 11. ML module (model selection).

In the Hyperparameter Tuning submodule (Figure 12), the user can adjust the search space for each machine learning model parameter. By default, WQEye provides a recommended search space as described in the paper. Users may also define their own parameter ranges if preferred. Additionally, the number of folds for cross-validation can be specified; in this example, five folds have been selected. Once the settings are confirmed, the user clicks the Tune button to begin hyperparameter tuning. The software tests all possible parameter combinations within the defined grid, and the best set of parameters is selected for model training.

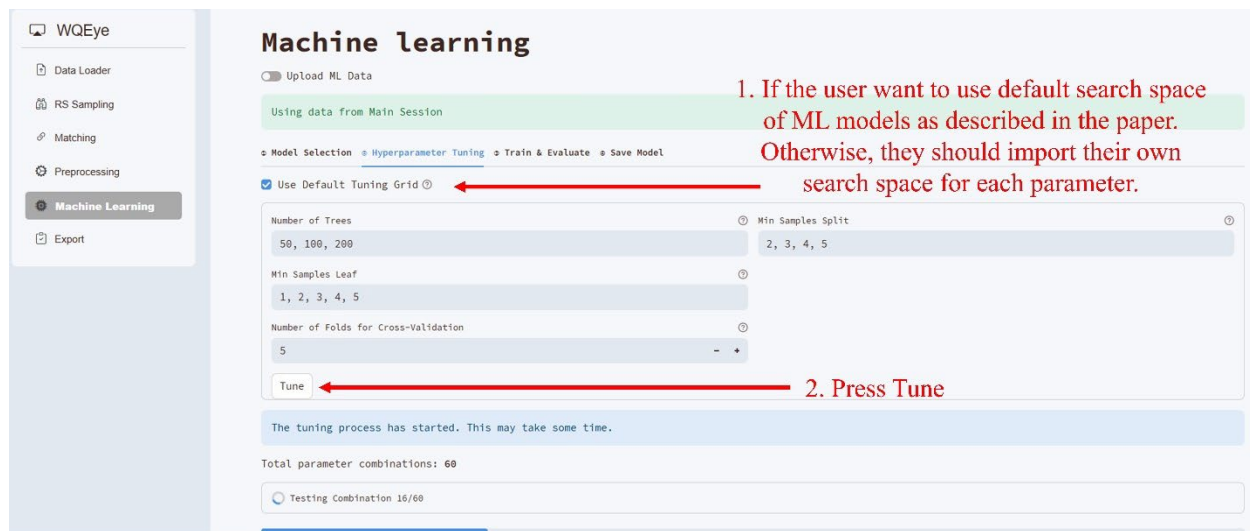


Figure 12. ML module (Hyperparameter Tuning).

In the Train and Evaluate submodule, the model is trained on the entire training dataset using the best hyperparameters identified during the tuning process. Once training is complete, the software automatically computes performance metrics.

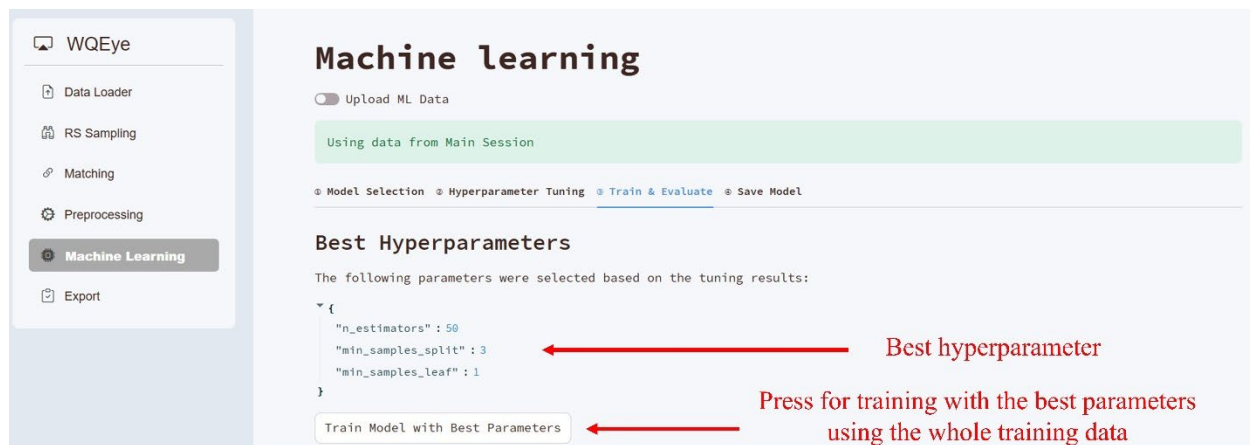


Figure 13. ML module (Training and evaluate).

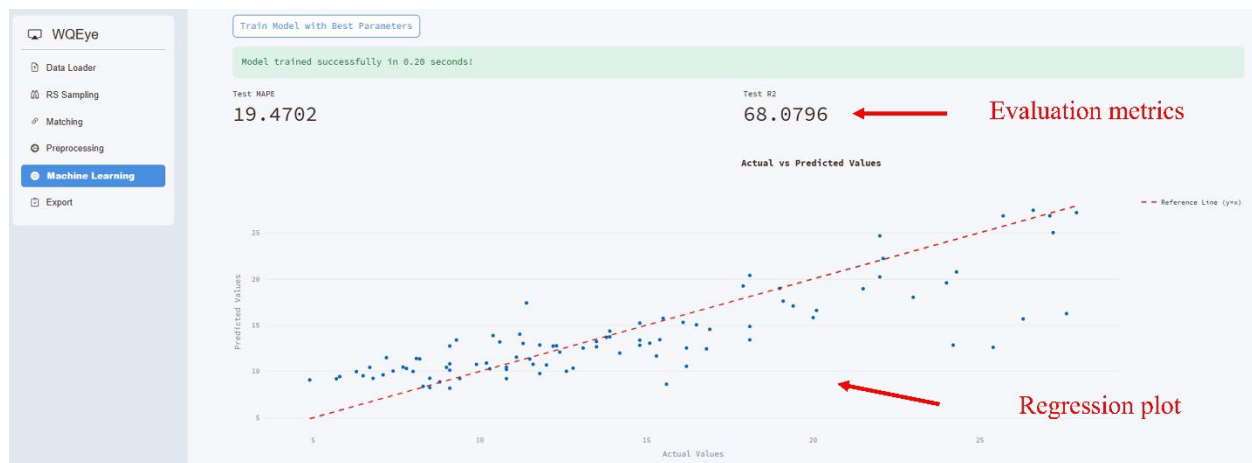


Figure 14. ML module (Training and evaluate).

Finally, in the Save Model submodule, the trained model can be saved for future use, allowing the user to apply it in other analyses or workflows as needed. The model is saved in a .zip file. For example, in the case of a Random Forest model, the .zip archive will contain:

- a model.pkl file, which is the trained model itself;
- a scaler.pkl file, which stores all scalers used during preprocessing so that the user can apply the same scaling to new data; and
- a metadata.json file, which includes information such as the selected spectral bands, best hyperparameters, and evaluation metrics from training and testing.

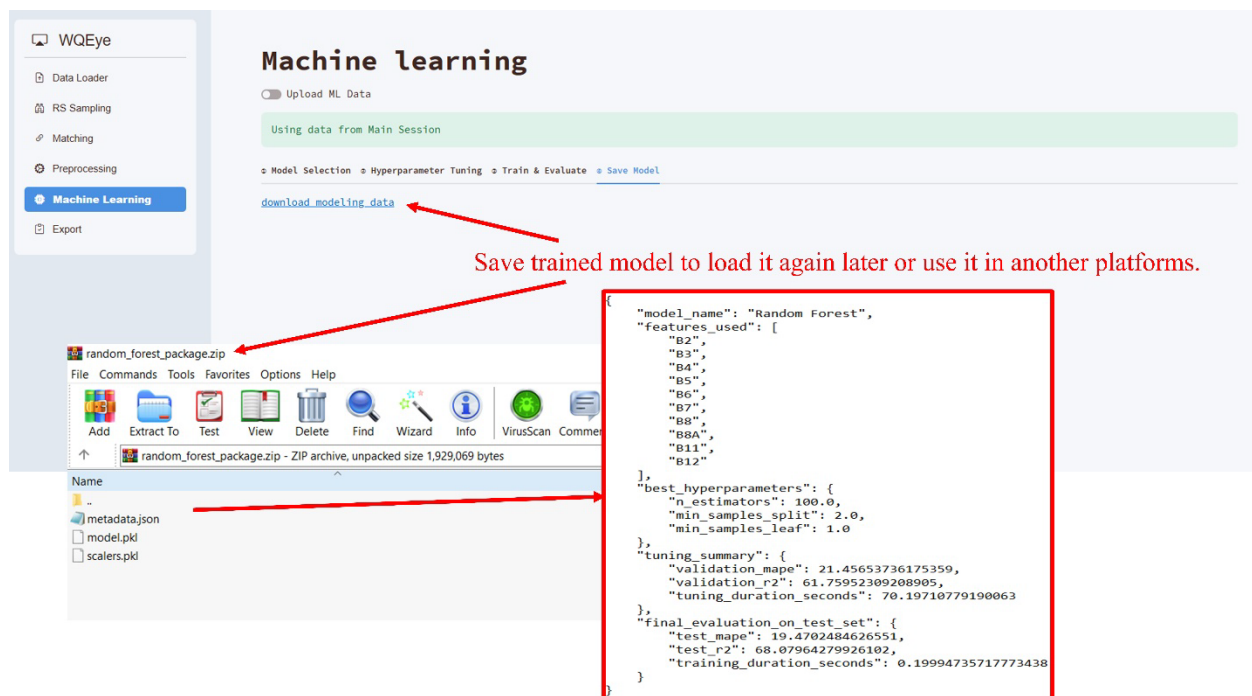


Figure 15. Saving trained model (the content of metadata.json file is also shown here).

1.7. Export module

In the final module (Export), the user can generate maps of the target water quality parameters using satellite imagery. The Export module contains three submodules. The first submodule allows the user to either draw or upload a region of interest (ROI). The user can upload a .geojson file directly or draw a polygon on the map, export it as a .geojson file, and then upload it to define the ROI.

Important: If the user draws a polygon, they must export and upload it again. Simply drawing the polygon is not sufficient to specify the region for processing.

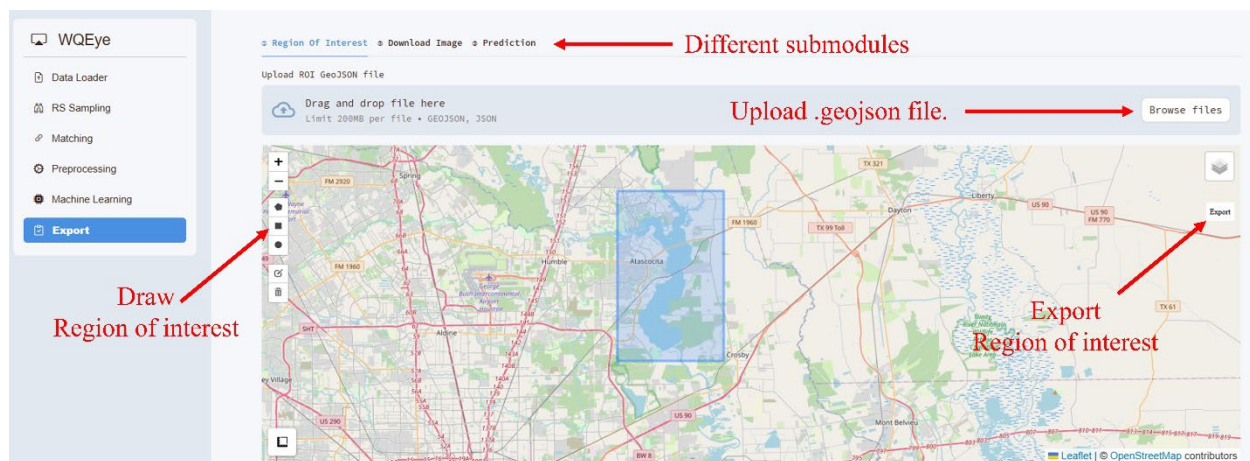


Figure 16. Export (specifying the region of interest).

Once the user uploads their region .geojson file, it will be displayed on the map with a dashed red boundary box.

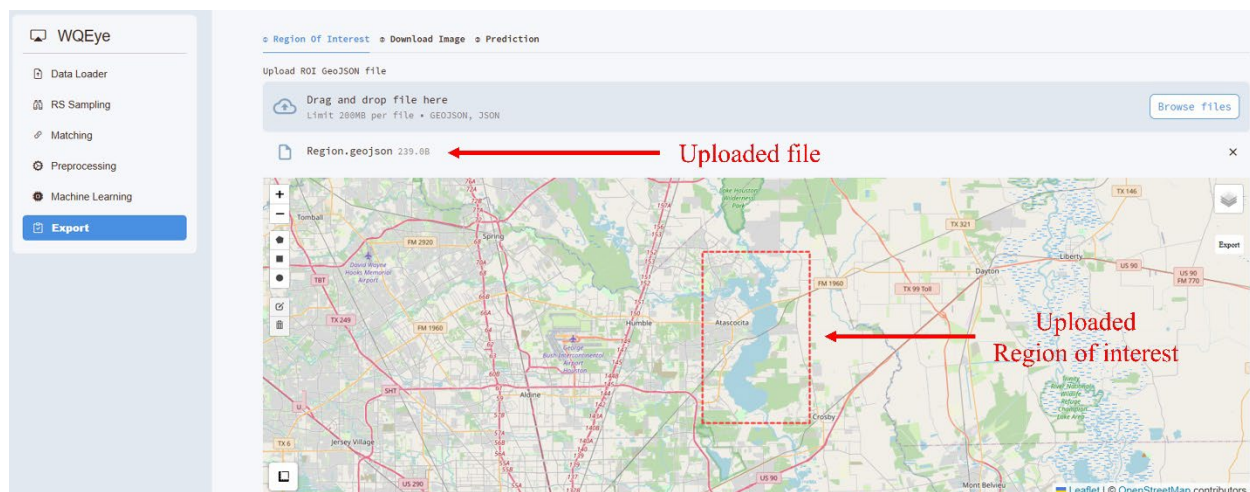


Figure 17. Export (specifying the region of interest).

Then, in the second submodule (Download Image), the user must provide a satellite image to be used for the estimation of the target parameter. At this stage, the user can select the desired satellite mission (Sentinel-2 or Landsat-8/9).

Note: The selected mission must match the one used to train the machine learning model.

The user can also specify a target date, based on which the system will retrieve the nearest available images with cloud coverage below a user-defined threshold (set within the submodule). Additionally, the user can choose the output resolution, which will determine the resolution of the final parameter map.

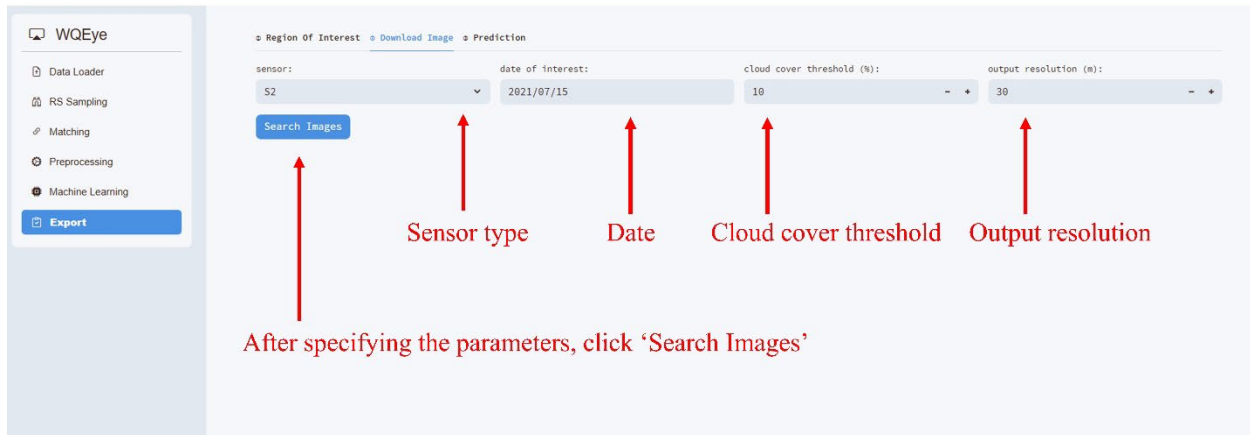


Figure 18. Export (specifying the image of interest).

Once the Search Images button is pressed, images matching the specified parameters within ± 1 month of the selected date will be displayed on the map. It should be noted that all images overlapping the region of interest (ROI) are shown, and the user can select the one they wish to use for generating the water quality map. The user can also view the water mask of the region (the generation procedure is fully explained in the paper). After selecting an image from the list below, the user can download it for further use. The downloaded file will contain two .tiff files: one covering the entire region and another covering only the water areas.

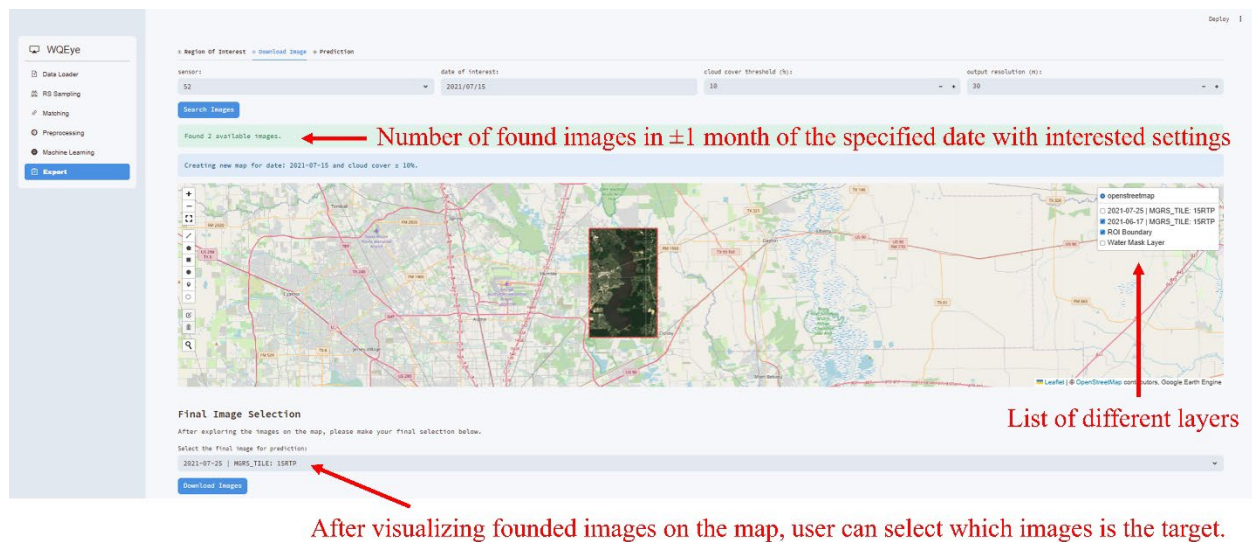


Figure 19. Export (specifying the image of interest).

As shown in Figure 20, the file names are generated based on the user-defined parameters and the specifications of the selected image.

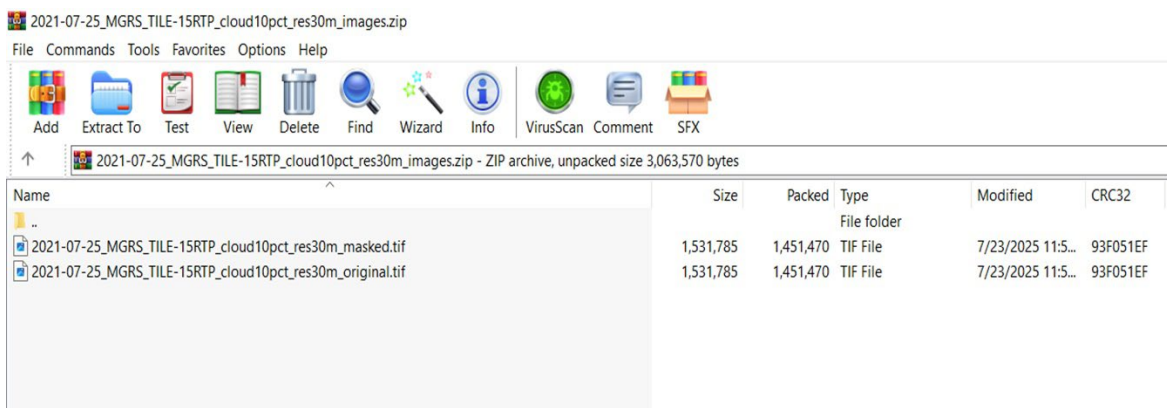


Figure 20. Downloaded image files.

In the Prediction submodule, the user can upload the .zip file containing the trained machine learning model (Figure 15) and click Run Prediction. The software will then apply the model to the satellite image selected in the previous step to predict the target water quality parameter. The resulting prediction map will be displayed and can also be downloaded for further use.

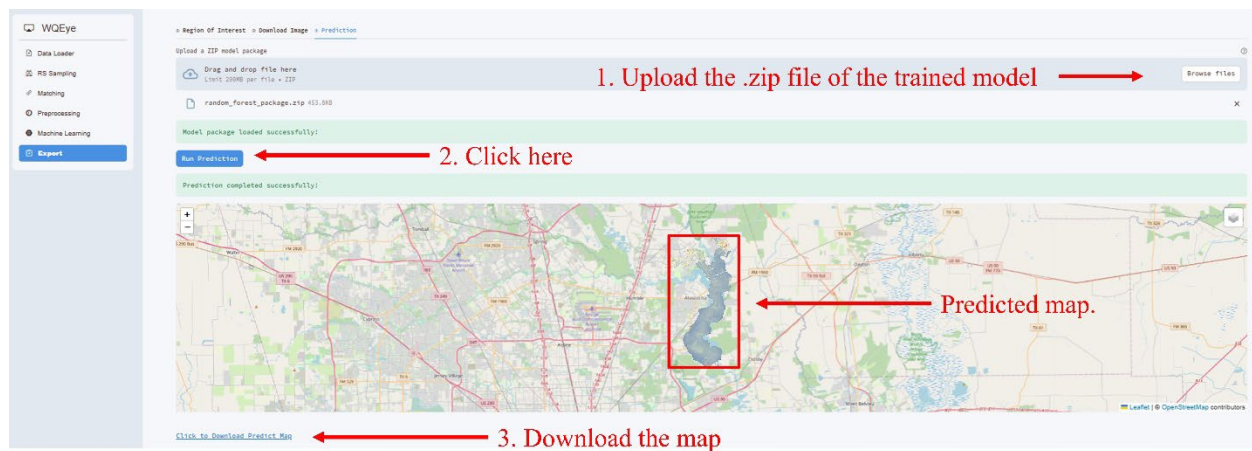


Figure 21. Generation of the water quality map.